ENR 1.7 ALTIMETER SETTING PROCEDURES

1 Introduction

The altimeter setting procedures in use conform to those contained in ICAO Doc 4444 ATM/501 Procedures for Air Navigation Services and ICAO Doc 8168 OPS/611 Procedures for Air Navigation Services - Aircraft Operations and are given in full below.

1.1 Flights required to comply with Altimeter setting procedures. These altimeter setting procedures shall be observed by all aircraft operating in level flight at or above 1 500 feet above the ground or water irrespective of the weather conditions and irrespective of whether the flight is operating on a VFR or IFR flight plan.

1.2 This does not imply that all cross country flights must be conducted at more than 1 500 feet above the ground or water. A pilot electing to conduct a flight in VMC at less than

1 500 feet above the ground or water may conduct the whole flight on QNH (or QFE) and at any altitude (or height) irrespective of the aircraft's heading.

2 Basic element of the system

2.1 System of flight levels.

a) During its en route flight an aircraft shall be flown along surfaces of constant atmospheric pressure called flight levels and related to an altimeter setting 1013,2 hPa. Throughout its en route flight an aircraft's vertical position shall be expressed in terms of flight levels.
b) Flight level zero shall be located at the atmospheric pressure level of 1013,2 hPa. Consecutive flight levels shall be separated by a pressure interval corresponding to 500 feet in the Standard Atmosphere.
c) Flight levels shall be numbered according to the following table, where the corresponding height in the Standard Atmosphere is specified in feet:

Flight level number	Altimeter Indication	
	Feet	Meters
15	1500	450
20	2000	600
50	5000	1500
100	10000	3050
150	15000	4550
200	20000	6100

3 Transition Altitude

a) During flight in the vicinity of an aerodrome at or below a fixed altitude called the transition altitude, an aircraft shall be flown at altitudes determined from an altimeter set to sea level pressure (QNH) and its vertical position shall be expressed in terms of altitude.

b) Transition altitudes are specified as follows for the aerodromes listed hereunder:

Aerodrome	Transition Altitude
ALEXANDER BAY	6000 FEET
BHISHO (Ciskei)	7000 FEET
BLOEMFONTEIN - BRAM FISCHER INTERNATIONAL AIRPORT	8000 FEET
CAPE TOWN INTERNATIONAL AIRPORT	7500 FEET
KING PHALO	7000 FEET
GEORGE	8000 FEET
GEORGE DICK MONTSHIOA AIRPORT	7000 FEET
GIYANI	5500 FEET
JOHANNESBURG (RAND)	8000 FEET
JOHANNESBURG - O R TAMBO INTERNATIONAL AIRPORT	8000 FEET
KIMBERLEY	7000 FEET
KLEINSEE	4500 FEET
LADYSMITH	9500 FEET
DURBAN - KING SHAKA INTERNATIONAL AIRPORT	5500 FEET
LANSERIA INTERNATIONAL AIRPORT	8000 FEET
LICHTENBURG	8000 FEET
LYDENBURG	7500 FEET
MALELANE	7500 FEET
MARGATE	5000 FEET
MESSINA	7000 FEET
KRUGER MPUMALANGA INTERNATIONAL AIRPORT	9000 FEET
NEWCASTLE	9500 FEET
PHALABORWA	3500 FEET
PIETERMARITZBURG (ORIBI)	8500 FEET
POLOKWANE INTERNATIONAL AIRPORT	9000 FEET
PIETERSBURG (CIVIL)	9000 FEET
PILANESBERG INTERNATIONAL AIRPORT	7400 FEET
PONGOLA	6500 FEET

Aerodrome	Transition Altitude
CHIEF DAWID STUURMAN INTERNATIONAL AIRPORT	5500 FEET
WONDERBOOM / PRETORIA	8000 FEET
PULLENSHOPE (Hendrina)	8000 FEET
RICHARDS BAY	3500 FEET
SECUNDA	8000 FEET
SISHEN	8000 FEET
SKUKUZA	9000 FEET
TSWALU	8000 FEET
TUTUKA POWER STATION	8000 FEET
TZANEEN	8500 FEET
ULUNDI	7100 FEET
MTHATHA	7500 FEET
UPINGTON INTERNATIONAL AIRPORT	5500 FEET
VENETIA MINE	7000 FEET
VEREENIGING	8000 FEET
WELKOM	7000 FEET

c) For all points of departure and arrival within 25 nautical miles of the aerodrome listed in paragraph 3 (b), the transition altitude shall be the same as that listed for such aerodromes.

d) In VMC, flights departing from or arriving at points beyond 25 nautical miles from any of the aerodromes tabled above, shall observe a height of 2 000 feet above the ground or water as the transition altitude.

e) In IMC, flights departing from or arriving at points beyond 25 nautical miles from any of the aerodromes tabled above, shall observe the lowest safe cruising altitude as the transition altitude.

4 Transition Levels

a) The change in reference from flight level used while enroute to altitude used in the vicinity of an aerodrome, shall be made at a horizontal plane located above the transition altitude called the transition level.

b) Transition levels vary with variations in barometric pressure in such a way that no transition level will be less than 1 000 feet above the transition altitude within 25 nautical miles of an aerodrome with an Air Traffic Service Unit (ATSU).

c) If an ATSU is in existence at the aerodromes tables above the current transition level for each aerodrome and for points within 25 nautical miles thereof shall be included in the approach and landing instructions.

d) In VMC, flights intending to land at points beyond 25 nautical miles from any of the aerodromes, listed in paragraph 3 (b), shall observe a height of 3 000 feet above the ground or water as the transition level.

e) In IMC, flights intending to land at points beyond 25 nautical miles from any of the aerodromes, listed in paragraph 3 (b), shall observe the flight level 500 feet above the lowest safe altitude as the transition level.

5 Application of procedure

5.1 Take-off and climb

a) A QNH altimeter setting shall be made available to aircraft in the routine take-off and climb clearance.

b) A QFE altimeter setting shall be made available to aircraft upon request.

c) At least one altimeter in an aircraft shall be set to the QNH before take-off and the reading thereon shall then be confirmed by comparison with the ground elevation. The vertical positioning of such aircraft during climb shall be by reference to altitude until the transition altitude is reached.

d) On reaching the transition altitude, at least one altimeter within the aircraft shall be set to 1013,2 hPa and thereafter the vertical position of such an aircraft shall be by reference to flight levels.

Note: On reaching the transition altitude, pilots shall reset their altimeters to 1013,2 hPa without requesting Air Traffic Control (ATC) permission to do so and without notifying Air Traffic Control (ATC) that the change has been made.

6 Enroute

6.1 Vertical Separation

a) The vertical separation during en route flight shall be expressed in terms of flight levels.

b) The vertical position of an aircraft shall be expressed in terms of flight levels both in AIR-REPS and in air traffic service messages.

6.2 Terrain Clearance

a) The Meteorological Forecast Office will, on request, make available to pilots the forecast lowest en-route QNH to enable pilots to determine the lowest flight level that will ensure adequate terrain clearance for routes or segments of routes on which this information is required.

b) The lowest safe flight level may be determined in one of the following ways:
- if there is more than one altimeter in the aircraft, set the subscale of one of them to the forecast lowest QNH, and then compare the readings of this altimeter with elevations shown on the map of the aircraft's route to ensure the minimum 1 500 feet terrain clearance, where applicable; or
- make a preflight check to ensure that the flight level selected will provide the minimum terrain clearance by determining the relationship which will exist between the forecast lowest QNH and the altimeter subscale setting (1013,2 hPa).

REMEMBER THE DANGER SIGNAL IS:
When the QNH is lower than the subscale setting the aircraft will be lower than indicated.

7 Approach and landing

a) A QNH altimeter setting shall be made available in routine approach and landing clearances.

b) A QFE altimeter setting shall be made available on request in approach and landing clearances.

c) The vertical positioning of aircraft during approach shall be controlled by reference to flight levels until the transition level is reached.

d) Vertical positioning of aircraft above the transition level may be by reference to altitude (QNH) or height (QFE), provided that, after the descent to land is commenced, level flight above the transition altitude is not indicated or expected.

Note: This exception to 7 (c) is intended to apply primarily to jet aircraft, for which uninterrupted descents from high altitudes are desirable.

e) On reaching the transition level at least one altimeter within the aircraft shall be set to the QNH and thereafter the vertical positioning of such aircraft shall be by reference to altitudes.

Note 1: On reaching the transition level, pilots shall reset their altimeters to the QNH without requesting Air Traffic Control (ATC) permission to do so and without notifying Air Traffic Control (ATC) that the change has been made.
Note 2: This does not preclude a pilot from using a QFE setting for terrain clearance purposes during the final approach to the runway.

f) During the approach to land, terrain clearance may be determined by using either a QNH altimeter setting (giving altitude) or a QFE setting (giving height above the QFE datum)

8 Missed approach

The relevant parts of paragraphs 2.1 and 4 shall apply in the case of a missed approach.

9 Description of altimeter setting region

a) ATSU will ensure that the latest QNH is always readily available for passing to an aircraft and for determining the current transition level.
b) ATSU will pass to aircraft the QFE on request.
c) Both the QNH and QFE are rounded down to the nearest whole hectopascal before being passed to the pilot. However, the QNH and QFE are available to the nearest tenth of a hectopascal if required.

10 Procedures applicable to operators (including pilots)

10.1 The levels at which the en-route phase of an IFR flight is to be conducted shall be specified in the flight plan in terms of flight levels. Flights conducted in the vicinity of an aerodrome shall be flown at altitudes if below the transition altitude and at flight levels if above the transition altitude.

10.2 No VFR flight shall be required to state its intended flight level/ altitude in a flight plan; Provided that, where a level is stated, it shall be -

a) In terms of flight levels if the flight is to be conducted at or above 1500 feet above the surface; or
b) In terms of altitudes if the flight is to be conducted at less than 1500 feet above the surface.

10.3 The flight level or levels selected for a flight shall -

a) In IMC ensure adequate terrain clearance;
b) For IFR flights, satisfy ATC clearances; and
c) For all IFR flights and for VFR flights at or above 1 500 feet above the surface, be compatible with the application of the semi-circular rule.

Note: Pilots must take note of the procedures applicable to VFR flights as described - see AIP ENR 1.2.1

10.4 Procedures applicable to ATSU

a) ATSU shall ensure that the latest QNH is always readily available for passing to aircraft and for determining the current transition level.
b) Both the QNH and the QFE shall be rounded down to the nearest whole hectopascal before being passed to pilots. However the QNH and QNE shall be available to the nearest tenth of a hectopascal, if required.

11 Semi-circular rule

a) Unless otherwise directed by an air traffic service unit, the pilot in- command of an aircraft in level flight shall fly at an appropriate flight level selected according to magnetic track from the table below: -

b) Aircraft flown in accordance with VFR at a height of less than 1500 feet above the surface, shall not be required to comply with para (a) above, unless if otherwise directed by an air traffic service unit.

c) A flight conducted from flight level 200 and above, shall be flown in compliance with IFR.

MAGNETIC TRACK			
Flight Level			
From 000° to 179°		**From 180° to 359°**	
IFR	**VFR**	**IFR**	**VFR**
	15	20	25
30	35	40	45
50	55	60	65
70	75	80	85
90	95	100	105
110	115	120	125
130	135	140	145
150	155	160	165
170	175	180	185
190	195	200	
210		220	
230		240	
250		260	
270		280	
290		300	
310		320	
330		340	
350		360	
370		380	
390		400	
410		430	
450		470	
490		510	
etc.		etc.	

d) Note: For further application of the semi-circular rule, refer to ICAO Annex 2, Appendix A and B.